

Multimodal Cancer Cell Classification

30 versions, six diagnosed failures, one breakthrough

A methodological run at the Multimodal Cancer Classification challenge — where each diagnosed failure shaped the next version, and the wins came from reading the losses faithfully.

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PUBLIC LB

0.8355

v47 noisy-student distillation

LIFT VS V19

+0.090

VERSIONS

30

BF · 128 × 128

FL · 128 × 128

Binary cell classification on paired BF + FL

Two structural facts shape every method choice that follows.

114,302

TRAIN CELLS · 12 patients

59,040

TEST CELLS · patient-disjoint

38.8%

POS RATE · pos_weight $\approx$ 1.58

0.8355

BEST LB · v47 distillation

- Per-cell prediction: is the cell from a cancer patient? (weak labels — MAC hypothesis, all cells inherit patient diagnosis)
- Paired inputs: brightfield + fluorescence, 128×128 grayscale
- 12 training patients · ~10k cells each — small N
- Strict patient-disjoint test set — OOD generalisation is the dominant failure mode
- Metric: AUC · 4 submissions/day · public/private LB split

**GRADING BASIS
(INSTRUCTOR, MAY 14)**
“What is done and how that is presented” — methodology over leaderboard score.

Dual EfficientNet-B0 with patient-MIL aux loss

The interventions that matter sit outside the backbone.

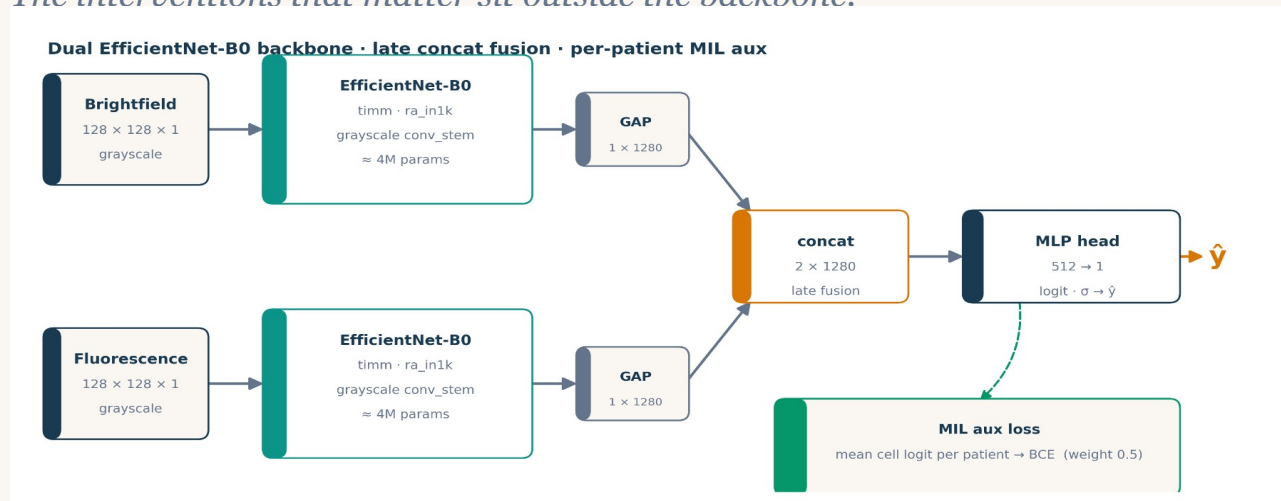


Fig. 1 Backbone is unremarkable on purpose. The interesting parts are at training time (MIL) and test time (AdaBN, stain norm).

- **Late concat fusion** *feature-level after GAP (v38 early fusion regressed -0.031)*
- **Per-patient MIL aux** *mean-pool cell logits per patient, BCE, $w=0.5$*
- **AdaBN** *refresh BN stats on test set (Li 2016)*
- **Test-set stain norm** *closes the 19.7 % FL mean gap*
- **8-way D4 TTA** *dihedral group, prob-averaged*

30 versions, 5 acts, one breakthrough

Each act answered a question. Each result changed the next question.



HEADLINE +0.081 LB total — the biggest single jump (+0.039) was soft-pseudo distillation (Hinton 2015).

v15 / v16: best CV I ever saw, worst LB

CoMIR-style cross-modal SSL — and a 29-point generalisation gap.

What we built

- Two ResNet-18 branches (BF + FL) with projection heads
- NT-Xent contrastive pretrain, $\tau = 0.1$ · 10 epochs
- Supervised fine-tune · 8 epochs · discriminative LR
- **3-fold stratified-group CV** *hid the failure — two of three folds had val patients similar to train*

DIAGNOSIS

Paired (BF, FL) of cell i always share a patient. Contrastive task solved at $\geq 95\%$ by encoding patient identity, not cell content.

0.866

CV AUC · 3-fold stratified-group · best I'd ever seen

0.572

PUBLIC LB · v15 submission · 29 AUC points of OOD gap

0.943 → 0.503

v16 RESPONSE · 9 careful fixes · LOPO CV — patient-OOF up, LB still random

v19: drop the SSL, model the test distribution

Five ingredients. None of them are new. Together: +0.174 LB over v15.

0.7455

PUBLIC LB · v19 · supervised floor · held ~3 weeks

+0.174

VS V15 SSL collapse → floor

19.7%

FL TEST brighter than train — why AdaBN + stain norm

Five ingredients in v19

- EfficientNet-B0 dual-branch (BF + FL), ImageNet-pretrained, late concat fusion
- Per-patient MIL auxiliary loss — patient-mean-logit BCE, weight 0.5
- AdaBN — refresh BN running stats on the test set (Li 2016)
- Test-set stain normalisation — pixel mean/std from test, not train
- Strong paired augmentation + 8-way D4 group TTA, prob-averaged

TAKE-HOME

The win wasn't a new architecture — it was treating the test distribution as the thing to model.

v41 vs v43: change one thing, not four

v41 · clean stack on top of v19

0.7563 **+0.011**

- label smoothing $\epsilon = 0.05$ *L4 standard*
- dropout **0.3** $\rightarrow$ **0.4** *classifier head*
- paired RRC *RandomResizedCrop scale 0.85-1.0*
- multiscale TTA *3 scales $\times$ 8 D4 = 24-way*

v43 · four MORE, stacked at once

0.7444 **-0.012**

- FL-tuned aug *CJ 0.5/0.3 + RandomGamma on FL only*
- WD bump *$1e-4 \rightarrow 3e-4$*
- 3-seed $\times$ SWA *Izmailov 2018 · last 4 epochs*
- 40-way TTA *5 scales $\times$ 8 D4*

RULE $\rightarrow$

One or two related variables per submitted version. v46 later reverted the FL-tuned aug + WD bump and recovered +0.079.

Use the test set itself as training signal

Pseudo-labels & iterative noisy student · v44 (hard) → v46 (soft) → v47 (round 2)

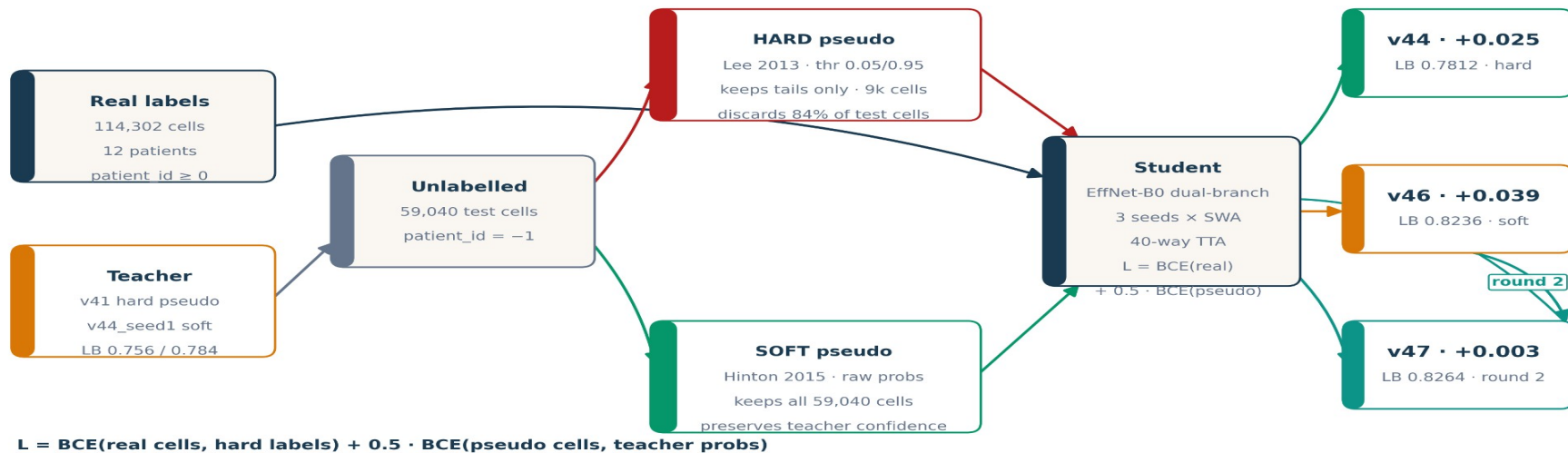


Fig. 2 Same teacher, two ways to use it. Hard discards 84 % of test cells. Soft keeps everything, weighted by teacher confidence. Round 2 uses v46 as the new teacher → v47.

Hard drops 84% of test cells; soft keeps all

Same teacher. Same predictions. Different way of using them.

METHOD	CELLS USED	Δ LB
baseline v41	114,302	—
v44 · hard	+ 9,350 (16 %)	+0.025
v46 · soft	+ 59,040 (100 %)	+0.039
v47 · round 2	+ 59,040 (100 %)	+0.003

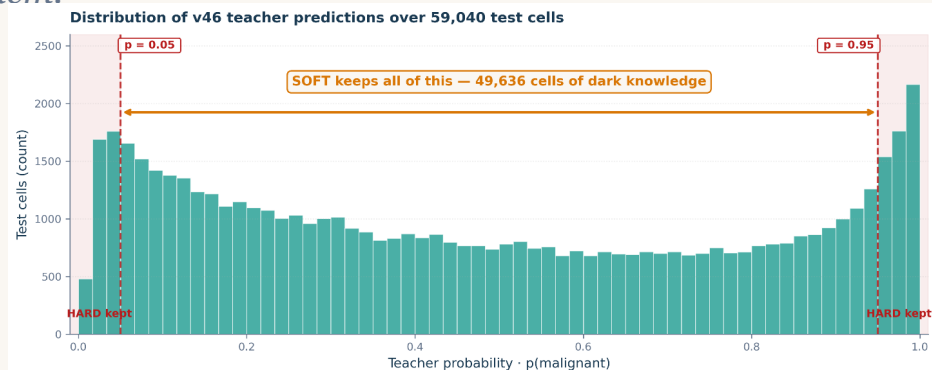


Fig. 3 Distribution of teacher predictions. Red dashed = hard thresholds. Soft keeps every column.

DARK KNOWLEDGE (HINTON 2015)

A cell labelled $p = 0.78$ by the teacher carries directional + magnitude information that hard binarisation discards. Soft targets preserve all of it.

WHY SOFT ALSO HELPS ROUND 2

v47's +0.003 lift on top of v46's +0.039 is small in mean — but per-seed tr_auc tightened (0.989-0.993 $\rightarrow$ 0.993-0.994). Round-2 acts as a variance reducer.

The whole story in one chart

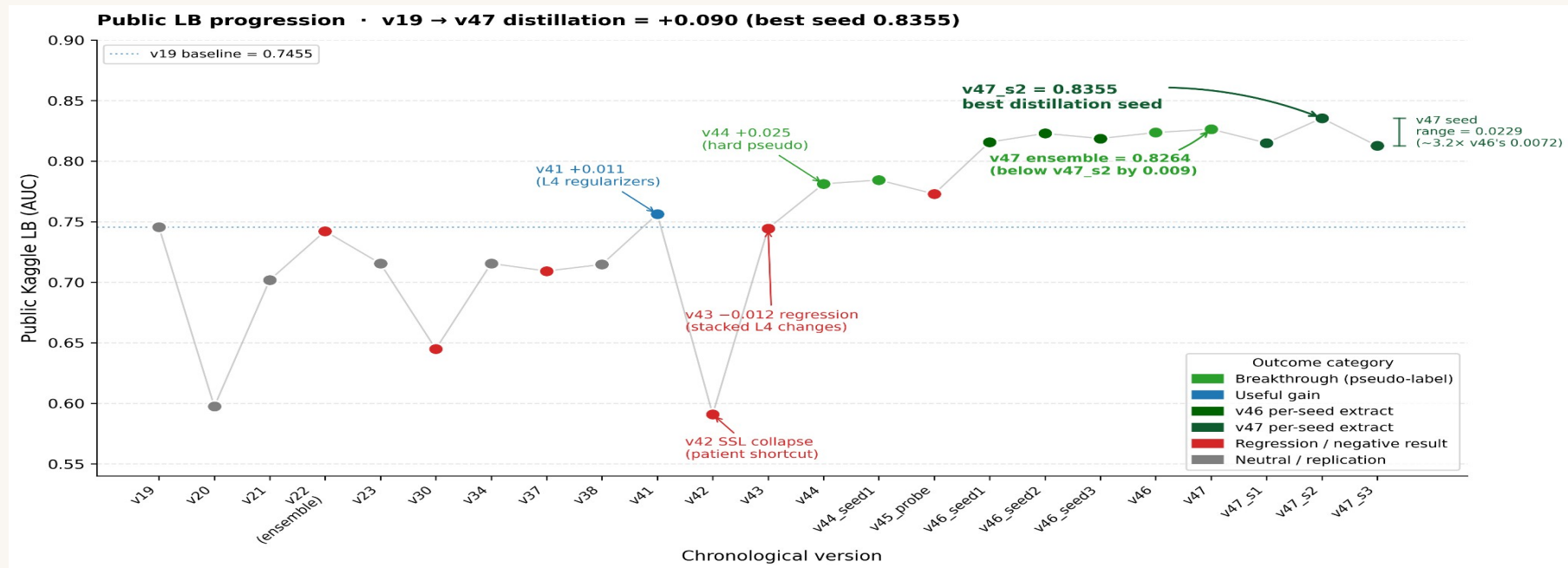


Fig. 4 Public LB across 30+ logged submissions. Green = pseudo-label breakthrough · blue = useful gain · red = regression · grey = neutral.

Six diagnosed failures that shaped the winning recipe

v22

Cross-recipe ensemble collapse

Sigmoid-avg of v19 + v21 at a 0.05 LB gap regressed below v19 alone (0.7422 vs 0.7455).

RULE → *Don't ensemble across recipes beyond a 0.02 LB gap.*

v27

Held-out val too noisy on N=12

2-patient holdout (pat_5 saturated FL, pat_14 dark FL) selected best_epoch = 0 — untrained.

RULE → *Use public LB as external validator; abandon held-out splits.*

v37 / v38

Lian 2024 transfer attempts

Heavy FL aug + early fusion both regressed (−0.036, −0.031). Lian's 4-ch ≠ our 1-ch grayscale.

RULE → *Re-validate any borrowed ablation on the actual modality.*

v42

SSL patient-shortcut collapse

CoMIR InfoNCE solved by patient-ID (loss = 0.02 in epoch 1). tr_auc 0.96 · LB 0.59.

RULE → *Cross-modal SSL amplifies patient-grouped shortcuts.*

v43

Stacked regulariser regression

Four simultaneous changes on top of v41 → −0.012 LB. No attribution possible.

RULE → *One or two related variables per submission.*

v45_probe

Ensemble rule confirmation

Sigmoid-avg of v41 + v44 (0.025 LB gap) regressed 0.008 below v44. Pearson 0.936.

RULE → *Tightened the rule from §22 to ≤ 0.015 LB gap.*

Ensembling never beat the best single seed

0.8264

3-seed average — BELOW the 0.8355 best single seed

0.7074

orthogonal feature-GBM blend — CV said 0.82, the LB said this

0.7422

cross-recipe average (v22) — below v19 (0.7455) alone

- **Within-recipe averaging regresses.** v47's 3-seed mean (0.8264) fell below its best seed (0.8355). Per-seed LB spans ~0.023 on N=12 — variance, not signal to average out.
- **Cross-recipe and orthogonal blends regress too.** v22 (0.7422) fell below v19 (0.7455); a decorrelated feature-GBM blend scored 0.7074 — its leave-patients-out CV (0.82) did not transfer to unseen patients.

TAKE-HOME

Averaging gave no lift here — only lower variance. So I locked two variance-reduced models, not the lucky single seed.

Clean, stable training — no divergence on 12 patients

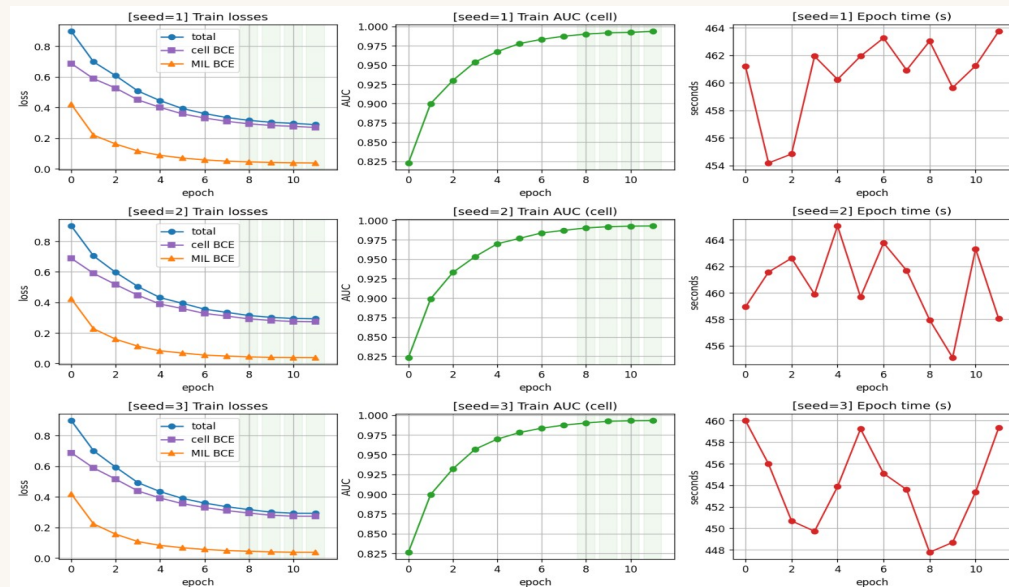


Fig. 5 v47 learning curves — per-cell train AUC + loss across seeds, 12 epochs. SWA over the last epochs.

What the curves show

- **Smooth convergence** train AUC rises to ~ 0.99 ; loss falls monotonically, no divergence
- **12 epochs** AdamW + OneCycleLR, batch 128, gradient clipping
- **SWA (Izmailov 2018)** weights averaged over the last epochs + a manual BN refresh
- **Per-patient MIL aux** regularises against within-patient memorisation
- **No held-out val** $N=12$ is too small; the public LB is the external check

Five things I'll carry forward

01**Higher CV $\neq$ higher LB.**

v15 had my best CV (0.866) and my worst LB (0.572). On patient-grouped OOD, CV alone is not a signal.

02**Don't stack unvalidated changes.**

v43 added four tweaks at once $\rightarrow$ -0.012 LB with no attribution. One or two related variables per experiment.

03**Seed variance is huge.**

Same recipe, different seed: 0.03 LB shift (v19 vs v23). Single-seed A/B claims under 0.03 are noise.

04**On small-N, data is the lever — not architecture.**

Four regularisers gained $+0.011$; soft-pseudo distillation gained $+0.042$ — about $4\times$ more lift from growing the effective dataset than from any architecture or regulariser change.

05**Negative results are the methodology.**

The v46/v47 winning recipe traces directly to six diagnosed failures. Reading the failures faithfully is what produced the win.